Hani Sabaie

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Education

Tabriz University of Medical Sciences **2018 - 2021**
Master of Science in Human Genetics *Tabriz, Iran*

- **GPA: 18.81/20**
- Thesis: Analysis of *BCAS4/hsa-miR-185-5p/SHISA7* Competing Endogenous RNA Axis in Late-onset Alzheimer's Disease, under the supervision of Dr. Maryam Rezazadeh; **Score: 20/20**
- Selected coursework: Population Genetics (**19/20**), Human Genetics (**18.25/20**), Molecular Genetics (**18.25/20**), Cytogenetics (**17.5/20**)

University of Guilan **2012 - 2017**
Bachelor of Science in Biology - Animal Sciences *Rasht, Iran*

- **GPA: 16.21/20**; GPA of the final two years: **17.74/20**
- Project: Genome Evolution, under the supervision of Dr. Shohreh Siyam; **Score: 19.25/20**
- Selected coursework: Biochemistry (**17/20**), Biostatistics (**19.5/20**), Cell and Molecular Biology (**20/20**), Molecular Genetics (**19.5/20**)

Research Interests

- Structural Variation
- Variant-to-Function Mapping
- Diagnostic Genetics and Genomics
- Statistical and Computational Genomics
- Functional Genomics and Gene Regulation
- Clinical Transcriptomics and RNA-seq
- Single-Cell Multiomics

Selected Publications

- Saeidian AH*, **Sabaie H***, Akbarzadeh M*, Vahidnezhad H, Youssefian L, Saadattalab T, Du H, Luo X, Owen NM, Zhao X, Glessner J, Hakonarson H. (2026). Computational strategies for copy number variation detection, disease association, and beyond. *Genome Biology*. 27(1):287. doi:10.1186/s13059-026-04189-6 (*co-first authors)

- Saed A, Akbarzadeh M, Gohari F, Asadrouh N, Jahromizadeh AM, **Sabaie H**, Zarkesh M, Hedayati M, Azizi F, Daneshpour MS. (2026). Serum ferritin and delirium risk: an integrative genomic analysis of causal inference and multi-tissue regulatory signals. *Human Genomics*. 20(1):97. doi:10.1186/s40246-026-00972-5

- **Sabaie H***, Taghavi Rad A*, Shabestari M, Habibi D, Saadattalab T, Seddiq S, Saeidian AH, Zahedi AS, Sanoie M, Vahidnezhad H, Zarkesh M, Foroutani L, Hakonarson H, Azizi F, Hedayati M, Daneshpour MS, Akbarzadeh M. (2025). Mitochondrial DNA copy number as a hidden player in the progression of multiple sclerosis: a bidirectional two-sample Mendelian randomization study. *Molecular Neurobiology*. 62(9):11643–11653. doi:10.1007/s12035-025-04980-9 (*equal contribution)

- Shoaran M, **Sabaie H**, Mostafavi M, Rezazadeh M. (2024). A comprehensive review of the applications of RNA sequencing in celiac disease research. *Gene*. 927:148681. doi:10.1016/j.gene.2024.148681

- **Sabaie H**, Talebi M, Gharesouran J, et al. (2022). Identification and analysis of *BCAS4/hsa-miR-185-5p/SHISA7* competing endogenous RNA axis in late-onset Alzheimer's disease using bioinformatic and experimental approaches. *Frontiers in Aging Neuroscience*. 14:812169. doi:10.3389/fnagi.2022.812169

- **Sabaie H**, Gharesouran J, Asadi MR, Farhang S, Ahangar NK, Brand S, Arsang-Jang S, Dastar S, Taheri M, Rezazadeh M. (2022). Downregulation of *miR-185* is a common pathogenic event in 22q11.2 deletion syndrome-related and idiopathic schizophrenia. *Metabolic Brain Disease*. 37(4):1175–1184. doi:10.1007/s11011-022-00918-5

- **Sabaie H**, Moghaddam MM, Moghaddam MM, Ahangar NK, Asadi MR, Hussen BM, Taheri M, Rezazadeh M. (2021). Bioinformatics analysis of long non-coding RNA-associated competing endogenous RNA network in schizophrenia. *Scientific Reports*. 11:24413. doi:10.1038/s41598-021-03993-3

Complete list of **30**+ peer-reviewed articles on Google Scholar.

Manuscripts Under Review

- Integrative single-nucleus multi-omics and Mendelian randomization analyses prioritize *SMAD3* as a candidate susceptibility gene linking aging-associated fibro-adipogenic progenitor dysregulation to inguinal hernia. *Human Genetics and Genomics Advances*, under review. Role: formal analysis and data curation (shared), original draft.
- Diagnostic transcriptomics in rare genetic disease: from variant interpretation to clinical decision-making. *Genetics in Medicine*, under review. Role: investigation, visualization, review and editing.

Research Experience

Independent Researcher **2025 - Present**

- Perform integrative **single-nucleus multiomics** (snRNA-seq, snATAC-seq) with Seurat, Signac, **hdWGCNA**, chromVAR and CellChat to study cell-type-specific gene regulation in complex disease
- Run **Mendelian randomization**, SMR/HEIDI, **colocalization**, GCTA-COJO and **SuSiE fine-mapping** on GWAS (UK Biobank, FinnGen) and eQTL (GTEx) data
- Co-first author of a **Genome Biology** review of germline **CNV detection**, benchmarking and disease association across SNP arrays, exome sequencing, genome sequencing and long-read sequencing, with the Center for Applied Genomics (CHOP) and Baylor College of Medicine
- Co-authored a review of **diagnostic transcriptomics** in rare genetic disease, covering its role in variant interpretation and clinical decision-making

Tabriz University of Medical Sciences **2024 - 2025**

Research Assistant, Principal Investigator: Dr. Maryam Rezazadeh *Tabriz, Iran*

- Applied systems biology, **RNA-seq** analysis and **WGCNA** to identify gene modules, hub genes and regulatory networks in neurological conditions
- Detected and interpreted variants from **exome sequencing** data
- Assisted the principal investigator with manuscript and **grant proposal** preparation
- Mentored **Master's, Ph.D. and medical students** conducting thesis research under Dr. Maryam Rezazadeh, providing guidance throughout their research projects

ValiAsr Hospital **2022 - 2023**

Researcher *Tehran, Iran*

- Integrated bioinformatic and experimental approaches to identify **peripheral blood biomarkers** for major depressive disorder

Tabriz University of Medical Sciences **2019 - 2021**

Student Research Assistant and M.Sc. Thesis Researcher, Supervisor: Dr. Maryam Rezazadeh *Tabriz, Iran*

- Identified **ceRNA regulatory axes** in neurogenetic disorders (Alzheimer's disease, schizophrenia, multiple sclerosis, autism) by reanalyzing public microarray datasets from GEO, leading to first-author publications in journals including **Scientific Reports** and **Frontiers in Aging Neuroscience**
- Measured expression of candidate genes by **PCR, RT-qPCR** and agarose gel electrophoresis, and analyzed the RT-qPCR data statistically
- Conducted **systematic and scoping reviews** on the genetics of human disease, with an emphasis on neurogenetic conditions

Teaching Experience

Tabriz University of Medical Sciences **2021**

Teaching Assistant, Microarray Data Analysis (Extracurricular Course) *Tabriz, Iran*

- Delivered a **one-semester extracurricular course** for approximately **seven Master's students** in the department, designed to help them incorporate bioinformatics analyses into their thesis research
- Taught **microarray data analysis** using the GEO2R web interface and R-based workflows with GEOquery; raw-data analysis for Affymetrix, Illumina and Agilent (single- and two-channel) platforms; statistical comparisons across more than two study groups; and practical data visualization and bioinformatics tools

Conference Presentations

- **Sabaie H**, Asadi MR, Salkhordeh Z, Rezazadeh M. Investigation of competing endogenous RNA regulation of *BCAS4* and *SHISA7* in tau pathology in Alzheimer's disease. Poster, First International and Tenth National Bioinformatics Conference of Iran, Kish Island, Iran, February 2022.

Technical Skills

- **Programming:** R (statistical modeling, visualization, R Markdown), Python, Linux/Unix, Bash, Git/GitHub
- **Statistical and Quantitative Analysis Software:** R, SPSS, GraphPad Prism, GenEx, Comprehensive Meta-Analysis
- **Statistical Genetics:** TwoSampleMR, SMR/HEIDI, MR-PRESSO, GCTA-COJO, coloc, SuSiE
- **Single-Cell Multiomics:** Drop-seq pipeline, Seurat, Signac, decontX, scDblFinder, Monocle, hdWGCNA, chromVAR, CellChat
- **Bulk RNA-seq and Microarray:** HISAT2, STAR, featureCounts, edgeR, DESeq2, limma, GEOquery, GEO2R, WGCNA
- **Variant Analysis:** FastQC, Trimmomatic, BWA, GATK HaplotypeCaller, ANNOVAR, ExomeDepth, CADD, ClinVar, gnomAD, ClinGen, Franklin, VarSome, ACMG/AMP guidelines
- **Networks:** Cytoscape, STRING, GeneMANIA, Enrichr, miRTarBase, LncBase, ENCORI
- **Wet Lab:** DNA extraction, RNA isolation, cDNA synthesis, PCR, RT-qPCR, primer design (Primer3, Primer-BLAST), reference gene validation (NormFinder, geNorm)

Selected Computational Training

- Summary-data-based Mendelian Randomization (**2025**); Mendelian Randomization, Advanced and Introductory R for Data Science, Linux Essentials (**2024**, online)
- R Programming, Basic Bioinformatics, Biostatistics (**2020**)

Languages and Test Scores

- **English:** Fluent | **TOEFL:** planned before February 2027
- **Persian:** Native

Honors and Awards

- **Selected Researcher**, Research Week, Clinical Research Development Unit, Tabriz Valiasr Hospital (2023)

Academic Service

- **Peer Review:** Reviewer for **24 journals**, including Scientific Reports, PLOS ONE, BMC Medical Genomics, Molecular Neurobiology, Molecular Therapy - Nucleic Acids, Functional & Integrative Genomics and Gene.
  Web of Science reviewer profile

Volunteer Academic Service

- **Research Mentoring (2022 - Present):** Provide ongoing, primarily online guidance to students and researchers from various universities on study design, research implementation and data analysis
- **Human Genetics M.Sc. Entrance Examination Advising (2018 - 2021):** Provided free guidance via Telegram to candidates preparing for the national Master's entrance examination in Human Genetics, covering study resources, study strategies and familiarity with the examination format

References

- **Dr. Maryam Rezazadeh**, Associate Professor (M.Sc. thesis supervisor), Department of Medical Genetics, Faculty of Medicine, Tabriz University of Medical Sciences, Tabriz, Iran. rezazadehm@tbzmed.ac.ir
- **Dr. Hakon Hakonarson**, Professor and Director (research mentor), Center for Applied Genomics, Children's Hospital of Philadelphia, Philadelphia, PA, USA. hakonarson@chop.edu
- **Dr. Amir Hossein Saeidian**, Assistant Professor and Director (research mentor), Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX, USA. amirhossein.saeidian@bcm.edu
- **Dr. Hassan Vahidnezhad,** (research collaborator), Associate Professor, Department of Pediatrics, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA. vahidnezhh@chop.edu